Netflix String Analysis

DS002 Project 2

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at least 3 str_*() functions (3 different str_*() functions) done at least 1 fct_*() function done - change rating from str to fct at least 3 regular expressions done at least 1 lookaround done at least 2 illustrative, well-labeled plots or tables a description of what insights can be gained from your plots and tables a reference / documentation of the data source (same idea as in Project 1).

```
library(tidyverse)
library(openintro)
```

Read Netflix data into Positron.

This Netflix data is from the Tidy Tuesday on 2021-04-20, obtained from Kaggle with credit to Shivam Bansal. The dataset was collected from Flixable, a third-party Netflix search engine. Data includes TV shows and movies added onto Netflix from 2019 onwards (although actual movie release year dates back to 1925), incldluing information such as show title, director, cast, country, release year, and rating.

Credits for data source: Github Link

```
# Read in data from Tidy Tuesday
netflix_titles <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytues

#Check first few rows
head(netflix_titles)
```

```
# A tibble: 6 x 12
  show_id type    title director    cast  country date_added release_year rating
  <chr>   <chr>   <chr> <chr>       <chr> <chr>   <chr>             <dbl> <chr>
1 s1      TV Show 3%    <NA>        João~ Brazil  August 14~         2020 TV-
MA
```

```
2 s2      Movie   7:19  Jorge Mich~ Demi~ Mexico  December ~         2016 TV-
MA
3 s3      Movie   23:59 Gilbert Ch~ Tedd~ Singap~ December ~         2011 R
4 s4      Movie   9     Shane Acker Elij~ United~ November ~         2009 PG-
13
5 s5      Movie   21    Robert Luk~ Jim ~ United~ January 1~         2008 PG-
13
6 s6      TV Show 46    Serdar Akar Erda~ Turkey  July 1, 2~         2016 TV-
MA
# i 3 more variables: duration <chr>, listed_in <chr>, description <chr>
```

Given the information available on the Netflix data set, here are some questions of interest that we can explore:

- What are the most common movie and TV show genres represented in Netflix?
- Out of the movies on Netflix, which is the most common rating (maturity) and is there a pattern among certain genres?
- How has the distribution of movie running times (mins) changed over the years, from 1925 to 2021?

Code to clean the data

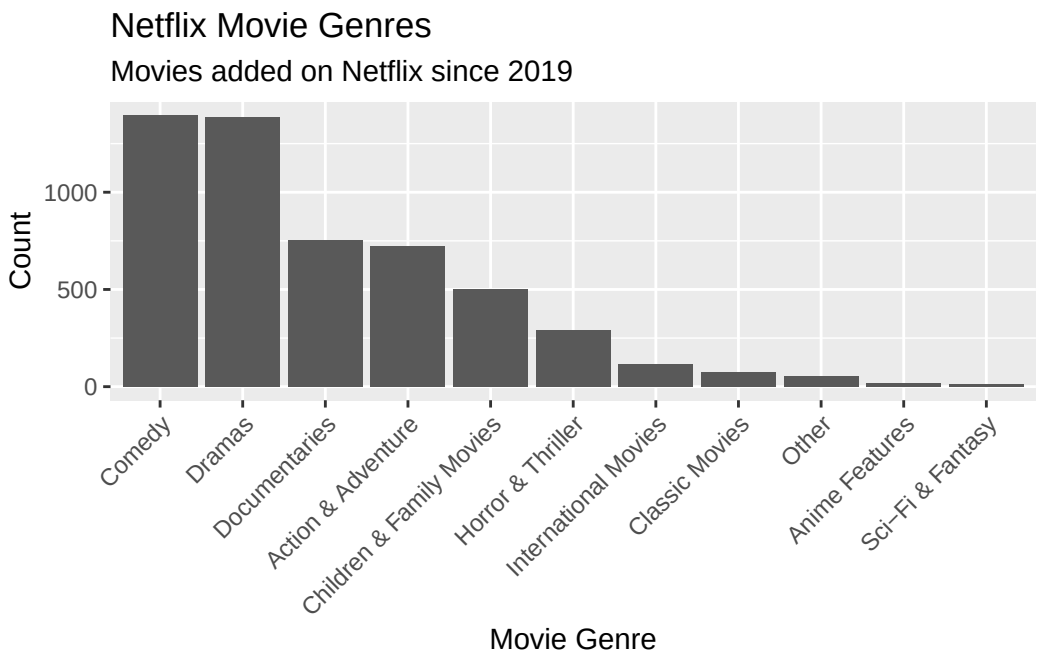
```
# Clean data to create new string variable that shows movie/ tv show genre

table(netflix_titles$rating)
```

```
       G    NC-17       NR       PG    PG-13        R    TV-14     TV-G
      39        3       84      247      386      665     1931      194
   TV-MA    TV-PG     TV-Y    TV-Y7 TV-Y7-FV       UR
    2863      806      280      271        6        5
```

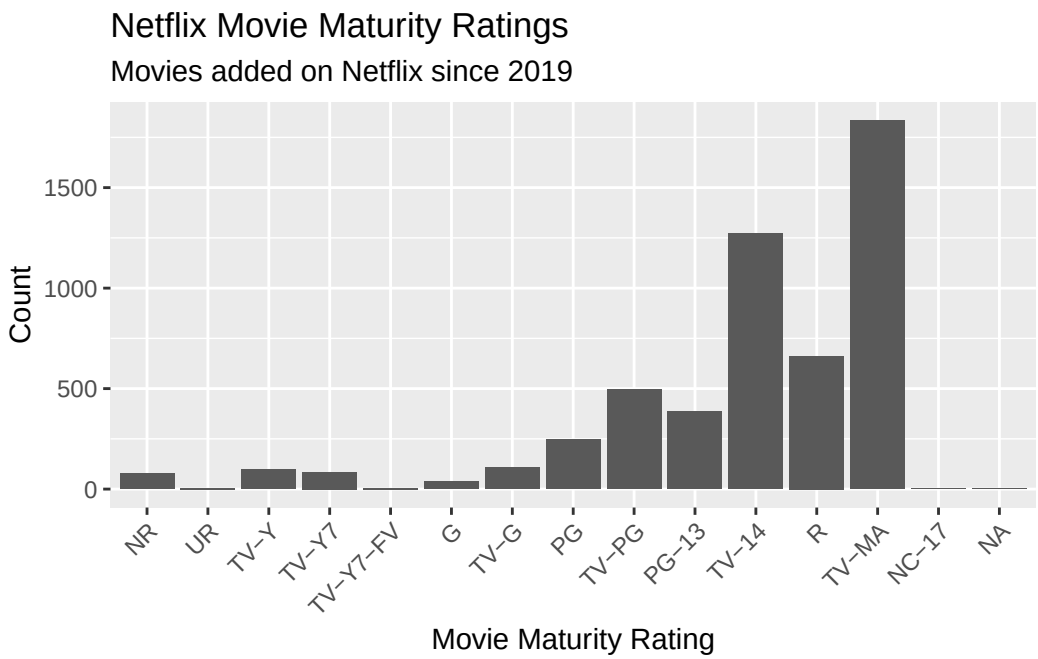
```
netflix_titles <- netflix_titles |>
  mutate(rating = fct_relevel(rating, c("NR", "UR", "TV-Y", "TV-Y7", "TV-Y7-FV", "G", "TV-G"

#NR = not rated
#UR = unrated
#TV-Y7-FV = a sub category of TV-Y7

netflix_titles <- netflix_titles |>
    mutate(genre = str_extract(listed_in,"^[^,]+"))|>
    mutate(genre2 = str_extract(listed_in,"(?<=, )[^,]+"))|>
    mutate(genre3 = str_match(listed_in,"^[^,]+, [^,]+, ([^,]+)")[, 2])
```

```r
# Create two separate data frames for movies only, because I want to analyse it separately
netflix_movies <- netflix_titles |>
  filter(type == "Movie")|>
  mutate(duration = as.integer(str_remove(duration," min$")))
```

```r
#Create a dataframe for only netflix movies

netflix_movies <- netflix_titles |>
  filter(type == "Movie")|>
  mutate(duration = as.integer(str_remove(duration," min$")))

#table(netflix_movies$genre)

#Group a few genres together that are more niche
netflix_movies <- netflix_movies |>
    mutate(genre_summary=fct_collapse(genre,
    "Horror & Thriller"= c("Horror Movies", "Thrillers"),
    Comedy= c("Stand-Up Comedy", "Comedies"),
    Other= c("LGBTQ Movies","Sports Movies","Music & Musicals","Romantic Movies","Independent
    "NA" = "Movies")) |>
        filter(genre_summary != "NA")

#Relevel factor by count

netflix_titles <- netflix_titles |>
  mutate(rating = fct_relevel(rating, c("NR", "UR", "TV-Y", "TV-Y7", "TV-Y7-FV", "G", "TV-G"

#table(netflix_movies$genre_summary)

#Create bar chart showing what movie genres are common
netflix_movies|>
    mutate(genre_summary = fct_infreq(genre_summary))|>
    ggplot(aes(x = genre_summary)) +
    geom_bar() +
    labs(
        x = "Movie Genre",
        y = "Count",
        title = "Netflix Movie Genres",
        subtitle = "Movies added on Netflix since 2019"
        ) +
            theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

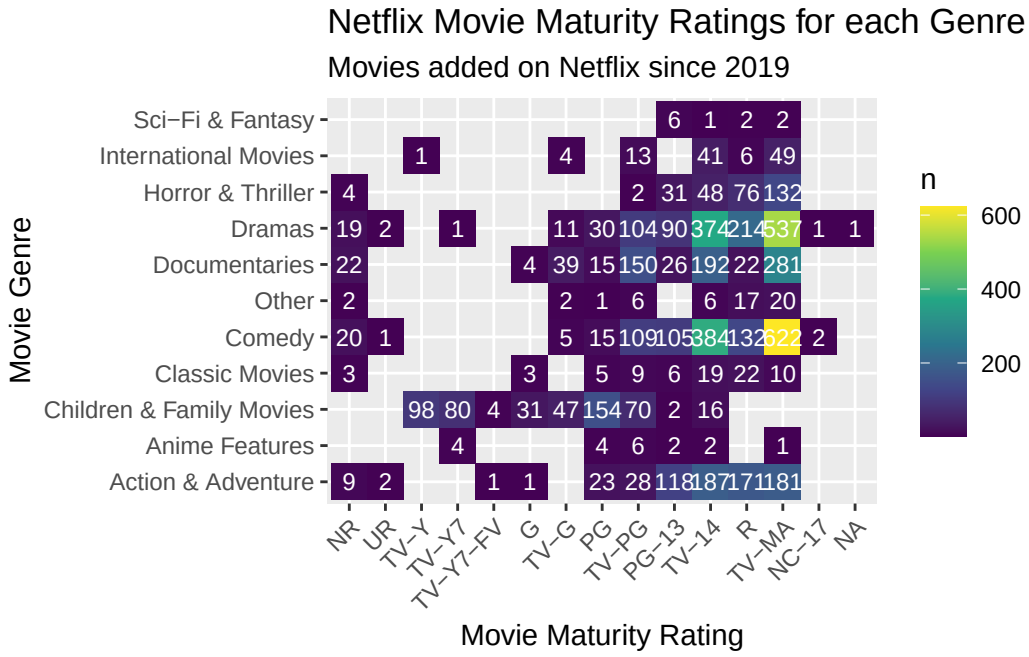



```
#Create bar chart showing what maturity ratings are common
netflix_movies|>
    ggplot(aes(x = rating)) +
    geom_bar() +
    labs(
        x = "Movie Maturity Rating",
        y = "Count",
        title = "Netflix Movie Maturity Ratings",
        subtitle = "Movies added on Netflix since 2019"
        ) +
            theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



The bar chart above displays the matury ratings of movies on Netflix (specifically movies added onto Netflix from 2019 onwards) from NR is "Not Rated" and UR is "Unrated," but other than those two ratings (refer to github source) for more information on each maturity rating

```
#Create heatmap showing what maturity ratings are common for what genre

netflix_movies |>
  count(genre_summary, rating) |>
  ggplot(aes(x = rating, y = genre_summary, fill = n)) +
  geom_tile() +
  geom_text(aes(label = n), color = "white", size = 3) +
    labs(
        x = "Movie Maturity Rating",
        y = "Movie Genre",
        title = "Netflix Movie Maturity Ratings for each Genre",
        subtitle = "Movies added on Netflix since 2019"
        ) +
  scale_fill_viridis_c() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
# Distribution of movie running times


#use date variable
#extract year
#change running time to numeric
```